

SECTION A (25 marks)

NO	ANSWER SCHEME	MARK	NOTES
1(a)	$\begin{aligned} \text{(i)} \quad \lim_{x \rightarrow 2} \sqrt{\frac{x^2 + 5x - 14}{x - 2}} &= \lim_{x \rightarrow 2} \sqrt{\frac{(x - 2)(x + 7)}{x - 2}} \\ &= \lim_{x \rightarrow 2} \sqrt{x + 7} \\ &= \sqrt{2 + 7} \\ &= 3 \end{aligned}$		Factorize and simplify Substitute $x = 2$
	$\begin{aligned} \text{(ii)} \quad \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 2}} &= \lim_{x \rightarrow \infty} \frac{\frac{x}{x}}{\sqrt{\frac{x^2}{x^2} + \frac{2}{x^2}}} \\ &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{2}{x^2}}} \\ &= \frac{1}{\sqrt{1 + 0}} \\ &= 1 \end{aligned}$		Divide each term by x $\lim_{x \rightarrow \infty} \frac{2}{x^2} = 0$
	$\begin{aligned} \text{(iii)} \quad \lim_{x \rightarrow -4^-} \frac{x^3 + 64}{ x + 4 } &= \lim_{x \rightarrow -4^-} \frac{(x + 4)(x^2 - 4x + 16)}{-(x + 4)} \\ &= \lim_{x \rightarrow -4^-} -(x^2 - 4x + 16) \\ &= -[(-4)^2 - 4(-4) + 16] \\ &= -48 \end{aligned}$		Factorize and simplify (must use $-(x + 4)$) Substitute $x = -4$
TOTAL MARKS		9 marks	

NO	ANSWER SCHEME	MARK	NOTES
2(a)	$f(x) = \frac{2}{3\sqrt{x}}$ $f'(x) = \lim_{h \rightarrow 0} \frac{\frac{2}{3\sqrt{x+h}} - \frac{2}{3\sqrt{x}}}{h}$ $= \lim_{h \rightarrow 0} \frac{2}{3h} \left(\frac{\sqrt{x} - \sqrt{x+h}}{\sqrt{x}(x+h)} \times \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}} \right)$ $= \lim_{h \rightarrow 0} \frac{2}{3h} \left(\frac{-h}{(\sqrt{x}(x+h))(\sqrt{x} + \sqrt{x+h})} \right)$ $= \frac{-2}{3(\sqrt{x^2})(2\sqrt{x})}$ $= -\frac{1}{3x^{\frac{3}{2}}}$		Correct First Principle Simplify and multiply with conjugate Try to solve
2(b)	$y = \ln \sqrt{\frac{1+e^x}{1-e^x}}$ $y = \ln \left(\frac{1+e^x}{1-e^x} \right)^{\frac{1}{2}}$ $y = \frac{1}{2} [\ln(1+e^x) - \ln(1-e^x)]$		Apply any suitable law of logarithms

NO	ANSWER SCHEME	MARK	NOTES
	$\frac{dy}{dx} = \frac{1}{2} \left(\frac{e^x}{1+e^x} + \frac{e^x}{1-e^x} \right)$ $= \frac{1}{2} \left[\frac{e^x(1-e^x) + e^x(1+e^x)}{(1+e^x)(1-e^x)} \right]$ $= \frac{1}{2} \left(\frac{e^x - e^{2x} + e^x + e^{2x}}{1 - e^{2x}} \right)$ $= \frac{1}{2} \left(\frac{2e^x}{1 - e^{2x}} \right)$ $= \frac{e^x}{1 - e^{2x}}$		Differentiate $\ln(1+e^x)$ Differentiate $\ln(1-e^x)$ Simplify In the simplest form
TOTAL MARKS		10 marks	

NO	ANSWER SCHEME	MARK	NOTES
3(a)	Total surface area = 216π $2\pi r^2 + 2\pi rh = 216\pi$ $r^2 + rh = 108$ $h = \frac{108 - r^2}{r}$ Volume, $V = \frac{2}{3}\pi r^3 + \pi r^2 h$ $V = \frac{2}{3}\pi r^3 + \pi r^2 \left(\frac{108 - r^2}{r} \right)$ $V = \frac{2}{3}\pi r^3 + 108\pi r - \pi r^3$ $V = 108\pi r - \frac{1}{3}\pi r^3$		Express h in terms of r Substitute h and simplify
3(b)	$V = 108\pi r - \frac{1}{3}\pi r^3$ $\frac{dV}{dr} = 108\pi - \frac{1}{3}\pi(3r^2) = 108\pi - \pi r^2$ $\frac{dV}{dr} = 0$ $108\pi - \pi r^2 = 0$ $r^2 = 108$ $r = \pm\sqrt{108}$ $r > 0, \therefore r = 6\sqrt{3}$		Find both $\frac{dV}{dr}$ and $\frac{d^2V}{dr^2}$ Find r when $\frac{dV}{dr} = 0$

NO	ANSWER SCHEME	MARK	NOTES
	$\frac{d^2V}{dr^2} = -2\pi r$ $\frac{d^2V}{dr^2} = -2\pi(6\sqrt{3}) = -12\sqrt{3}\pi = -65.3 < 0$ Since $\frac{d^2V}{dr^2} < 0$, then $r = 6\sqrt{3}$		Correct value of r with second derivative test (must $\frac{d^2V}{dx^2} < 0$)
TOTAL MARKS		6 marks	

SECTION B (75 marks)

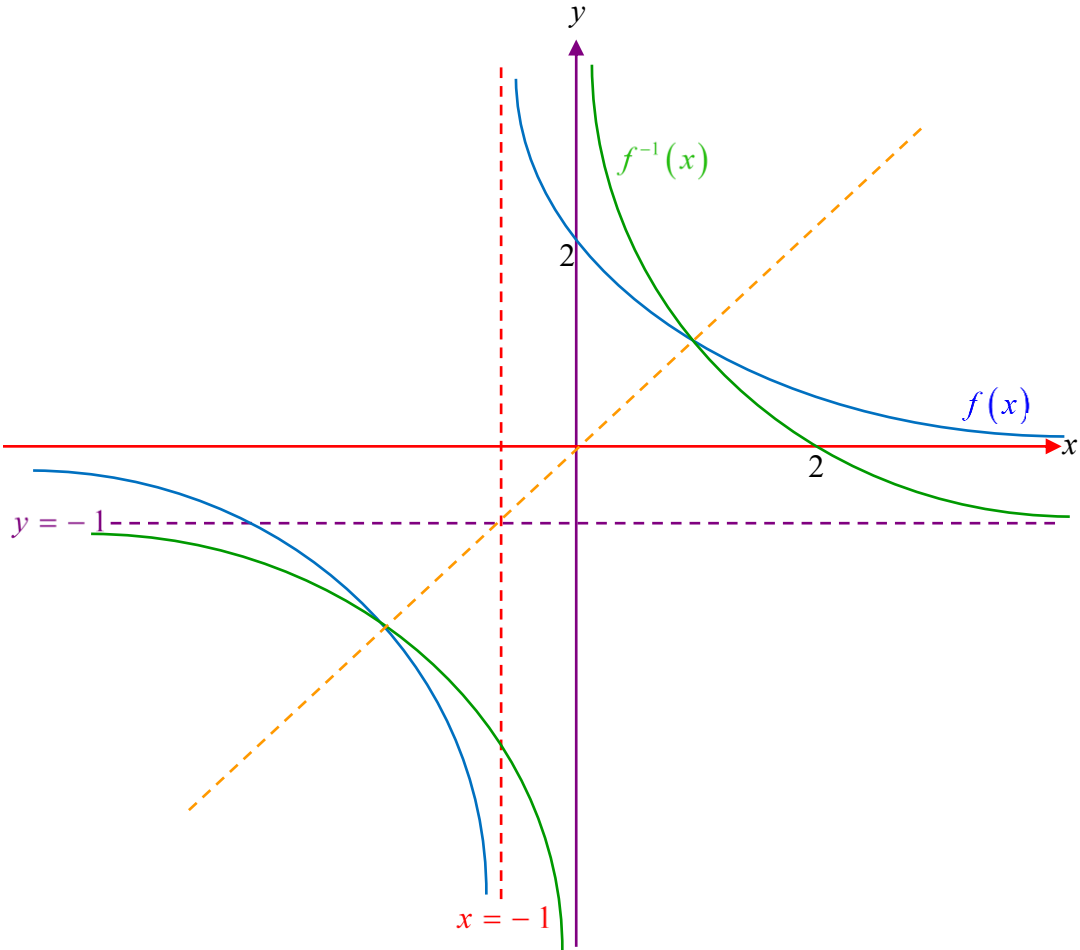
NO	ANSWER SCHEME	MARK	NOTES
1	$\frac{x}{2-i} + yi = (4-3i)2i^2$ $\frac{x}{2-i} \times \frac{2+i}{2+i} + yi = 8i^2 - 6i^3$ $\frac{2x + xi}{4-i^2} + yi = -8 + 6i$ $\frac{2x + xi}{5} + yi = -8 + 6i$ $2x + (x + 5y)i = -40 + 30i$ $2x = -40$ $x = -20$ $x + 5y = 30$ $5y = 50$ $y = 10$ $\therefore z = y + xi = 10 - 20i$		Simplify Solve x and y Correct x Correct y Correct z
TOTAL MARKS		5 marks	

NO	ANSWER SCHEME	MARK	NOTES																
2(a)	$\frac{x}{5+x} < \frac{1}{4}$ $\frac{x}{5+x} - \frac{1}{4} < 0$ $\frac{4x - 5 - x}{4(5+x)} < 0$ $\frac{3x - 5}{5+x} < 0$ <table border="1" data-bbox="322 651 1115 922"> <tr> <td>Interval</td><td>$(-\infty, -5)$</td><td>$\left(-5, \frac{5}{3}\right)$</td><td>$\left(\frac{5}{3}, \infty\right)$</td></tr> <tr> <td>$3x - 5$</td><td>—</td><td>—</td><td>+</td></tr> <tr> <td>$5 + x$</td><td>—</td><td>+</td><td>+</td></tr> <tr> <td>$\frac{3x - 5}{5 + x}$</td><td>+</td><td>—</td><td>+</td></tr> </table> $\left\{x : -5 < x < \frac{5}{3}\right\}$	Interval	$(-\infty, -5)$	$\left(-5, \frac{5}{3}\right)$	$\left(\frac{5}{3}, \infty\right)$	$3x - 5$	—	—	+	$5 + x$	—	+	+	$\frac{3x - 5}{5 + x}$	+	—	+		Try to get common denominator Any method
Interval	$(-\infty, -5)$	$\left(-5, \frac{5}{3}\right)$	$\left(\frac{5}{3}, \infty\right)$																
$3x - 5$	—	—	+																
$5 + x$	—	+	+																
$\frac{3x - 5}{5 + x}$	+	—	+																
2(b)	$\frac{3p^x - p^{-x}}{p^x + 4p^{-x}} = 2$ $\frac{3p^x - \frac{1}{p^x}}{p^x + \frac{4}{p^x}} = 2$ $\frac{3p^{2x} - 1}{p^{2x} + 4} = 2$																		

NO	ANSWER SCHEME		MARK	NOTES
	$3p^{2x} - 1 = 2(p^{2x} + 4)$ $p^{2x} - 9 = 0$ $(p^x - 3)(p^x + 3) = 0$ $p^x = 3$, $p^x = -3$ (ignored) $x \log p = \log 3$ $x = \frac{\log 3}{\log p}$	$x \ln p = \ln 3$ $x = \frac{\ln 3}{\ln p}$		Get quadratic equation Values of p^x Try to solve (taking log / ln both sides)
	Alternative $3p^{2x} - 1 = 2(p^{2x} + 4)$ $p^{2x} - 9 = 0$ $(p^x - 3)(p^x + 3) = 0$ $p^x = 3$, $p^x = -3$ (ignored) $x = \log_p 3$			Get quadratic equation Values of p^x Try to solve (taking log both sides)
TOTAL MARKS			11 marks	

NO	ANSWER SCHEME	MARK	NOTES
3(a)	$f(x) = \sqrt{a + bx}$ $f[f^{-1}(x)] = x$ $\sqrt{a + b f^{-1}(x)} = x$ $a + b f^{-1}(x) = x^2$ $f^{-1}(x) = \frac{1}{b}(x^2 - a)$ $f^{-1}(1) = 4, b = -2$ $f^{-1}(1) = -\frac{1}{2}(1^2 - a) = 4$ $1 - a = -8$ $a = 9$		Try to find $f^{-1}(x)$ Substitute $f^{-1}(1) = 4, b = -2$ into his $f^{-1}(x)$
3(b)	$f(x) = 5 - e^{-x}, x \in \mathbb{R}, g(x) = \ln(x - 4), x > 4$ $(f \circ g)(x) = 5 - e^{-\ln(x-4)}$ $= 5 - e^{\ln(x-4)^{-1}}$ $= 5 - \frac{1}{x-4}$ $= \frac{5x-21}{x-4}$ $\sqrt{(f \circ g)(5)} = \sqrt{\frac{25-21}{5-4}} = 2$		
TOTAL MARKS		8 marks	

NO	ANSWER SCHEME	MARK	NOTES
4(a)	$f(x) = \frac{2}{x+1}$ Domain : $x+1 \neq 0$ $x \neq -1$ $D_f = (-\infty, -1) \cup (-1, \infty)$ Range : $y \neq 0$ $R_f = (-\infty, 0) \cup (0, \infty)$		Find domain & range
4(b)	$f(x_1) = f(x_2)$ $\frac{2}{x_1+1} = \frac{2}{x_2+1}$ $x_2+1 = x_1+1$ $x_2 = x_1$ Since $f(x_1) = f(x_2)$ implies $x_1 = x_2$, then $f(x)$ is a one – to – one function.		
4(c)	$f[f^{-1}(x)] = x$ $\frac{2}{f^{-1}(x)+1} = x$ $f^{-1}(x)+1 = \frac{2}{x}$ $f^{-1}(x) = \frac{2}{x} - 1$ $D_{f^{-1}} = (-\infty, 0) \cup (0, \infty)$		

NO	ANSWER SCHEME	MARK	NOTES
4(d)			Shape $f(x)$ (Quadrant I & III) Shape $f^{-1}(x)$ (Quadrant I & III) All correct (Asymptotes, x & y-intercepts, label)
	TOTAL MARKS	12 marks	

NO	ANSWER SCHEME	MARK	NOTES
5(a)	$\lim_{x \rightarrow 9} \frac{k}{\sqrt{x-5}} - \frac{2}{\sqrt{x}} = \frac{1}{6}$ $\frac{k}{\sqrt{9-5}} - \frac{2}{\sqrt{9}} = \frac{1}{6}$ $\frac{k}{2} = \frac{1}{6} + \frac{2}{3}$ $k = \frac{5}{6} \times 2$ $k = \frac{5}{3}$		Substitute $x = 9$ Attempt to get k
5(b)	(i) $\lim_{x \rightarrow 1} f(x)$ exist $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$ $\lim_{x \rightarrow 1^-} \frac{4x-4}{2x^2-3x+1} = \lim_{x \rightarrow 1^+} \frac{-(1-x)}{x-1} + n$ $\lim_{x \rightarrow 1^-} \frac{4(x-1)}{(2x-1)(x-1)} = \lim_{x \rightarrow 1^+} \frac{x-1}{x-1} + n$ $\lim_{x \rightarrow 1^-} \frac{4}{2x-1} = \lim_{x \rightarrow 1^+} 1 + n$ $4 = 1 + n$ $n = 3$		Must correct function for left and right limit Simplify and substitute $x = 1$
5(b)	(ii) $f(x)$ is continuous at $x = 1$ $\lim_{x \rightarrow 1} f(x) = f(1)$		

NO	ANSWER SCHEME	MARK	NOTES
	$4 = m^2 - (1 + 2)^2$ $m^2 = 4 + 9$ $m = \pm\sqrt{13}$ $m > 0, \therefore m = \sqrt{13}$		Try to get m
5(c)	Vertical Asymptote: $x^2 - 4 = 0$ $x = -2, x = 2$ $\lim_{x \rightarrow -2^-} \frac{2x}{x^2 - 4} = -\infty \quad / \quad \lim_{x \rightarrow -2^+} \frac{2x}{x^2 - 4} = +\infty$ $\lim_{x \rightarrow 2^-} \frac{2x}{x^2 - 4} = -\infty \quad / \quad \lim_{x \rightarrow 2^+} \frac{2x}{x^2 - 4} = +\infty$ The vertical asymptotes are $x = -2$ and $x = 2$. Horizontal Asymptote: $\lim_{x \rightarrow \infty} \frac{2x}{x^2 - 4} = \lim_{x \rightarrow \infty} \frac{\frac{2x}{x^2}}{\frac{x^2}{x^2} - \frac{4}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{2}{x}}{1 - \frac{4}{x^2}} = \frac{0}{1} = 0$ $\lim_{x \rightarrow -\infty} \frac{2x}{x^2 - 4} = \lim_{x \rightarrow -\infty} \frac{\frac{2}{x}}{1 - \frac{4}{x^2}} = \frac{0}{1} = 0$ Therefore $y = 0$ is a horizontal asymptote.		Either / or limits for $x \rightarrow -2^\pm$ Either / or limits for $x \rightarrow 2^\pm$
TOTAL MARKS		14 marks	

NO	ANSWER SCHEME	MARK	NOTES
6(a)	(i) $\frac{d}{dx}(1+x^2)(1+y^2) = \frac{d}{dx}(x^2-1)$ $(1+x^2)\left(2y\frac{dy}{dx}\right) + (1+y^2)2x = 2x$ $\frac{dy}{dx} = \frac{-2xy^2}{2y(1+x^2)}$ $\frac{dy}{dx} = -\frac{xy}{1+x^2}$		Correct Product Rule
	Alternative (i) $(1+x^2)(1+y^2) = x^2-1$ $\frac{d}{dx}(1+x^2+y^2+x^2y^2) = \frac{d}{dx}(x^2-1)$ $2x + 2y\frac{dy}{dx} + 2xy^2 + 2x^2y\frac{dy}{dx} = 2x$ $(1+x^2)2y\frac{dy}{dx} = -2xy^2$ $\frac{dy}{dx} = -\frac{xy}{1+x^2}$		
6(a)	(ii) $\frac{d}{dx}\left[(1+x^2)\frac{dy}{dx}\right] = \frac{d}{dx}(-xy)$ $(1+x^2)\frac{d^2y}{dx^2} + \frac{dy}{dx}(2x) = -x\frac{dy}{dx} + y(-1)$ $(1+x^2)\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0$		
	Alternative		

NO	ANSWER SCHEME	MARK	NOTES
	(ii) $\frac{d^2y}{dx^2} = - \left[\frac{(1+x^2) \left(y + x \frac{dy}{dx} \right) - xy(2x)}{(1+x^2)^2} \right]$ $\frac{d^2y}{dx^2} = \frac{2x^2y - y}{(1+x^2)^2}$ $(1+x^2) \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + y$ $= (1+x^2) \left[\frac{2x^2y - y}{(1+x^2)^2} \right] + 3x \left(-\frac{xy}{1+x^2} \right) + y$ $= \frac{2x^2y - y - 3x^2y + y + x^2y}{1+x^2}$ $= 0$		Find $\frac{d^2y}{dx^2}$ and substitute into equation
6(b)	(i) $x(t) = \sin(2\sqrt{t})$ $y(t) = \cos^2 \sqrt{t}$ $\frac{dx}{dt} = \cos(2\sqrt{t}) \times \left[2 \left(\frac{1}{2\sqrt{t}} \right) \right]$ $\frac{dy}{dt} = (2 \cos \sqrt{t}) (-\sin \sqrt{t}) \left(\frac{1}{2\sqrt{t}} \right)$ $\frac{dx}{dt} = \frac{\cos(2\sqrt{t})}{\sqrt{t}}$ $\frac{dy}{dt} = -\frac{\cos \sqrt{t} \sin \sqrt{t}}{\sqrt{t}}$		Differentiate $\frac{dx}{dt}$ Differentiate $\frac{dy}{dt}$ (at least 2 items correct)

NO	ANSWER SCHEME	MARK	NOTES
	$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$ $\frac{dy}{dx} = \left(-\frac{\cos \sqrt{t} \sin \sqrt{t}}{\sqrt{t}} \right) \left(\frac{\sqrt{t}}{\cos(2\sqrt{t})} \right)$ $\frac{dy}{dx} = -\frac{\sin(2\sqrt{t})}{2 \cos(2\sqrt{t})}$ $\frac{dy}{dx} = -\frac{1}{2} \tan(2\sqrt{t})$		Use $\sin(2\sqrt{t}) = 2 \sin \sqrt{t} \cos \sqrt{t}$
	Alternative (i) $x(t) = \sin(2\sqrt{t})$ $\frac{dx}{dt} = \cos(2\sqrt{t}) \times \left[2 \left(\frac{1}{2\sqrt{t}} \right) \right]$ $\frac{dx}{dt} = \frac{\cos(2\sqrt{t})}{\sqrt{t}}$ $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$	$y(t) = \cos^2 \sqrt{t}$ $\frac{dy}{dt} = (2 \cos \sqrt{t})(-\sin \sqrt{t}) \left(\frac{1}{2\sqrt{t}} \right)$ $\frac{dy}{dt} = -\frac{\cos \sqrt{t} \sin \sqrt{t}}{\sqrt{t}}$ $\frac{dy}{dt} = -\frac{\sin 2\sqrt{t}}{2\sqrt{t}}$	Differentiate $\frac{dx}{dt}$ Differentiate $\frac{dy}{dt}$ (at least 2 items correct) Use $\sin(2\sqrt{t}) = 2 \sin \sqrt{t} \cos \sqrt{t}$

NO	ANSWER SCHEME	MARK	NOTES
	$\frac{dy}{dx} = \left(-\frac{\sin 2\sqrt{t}}{2\sqrt{t}} \right) \left(\frac{\sqrt{t}}{\cos(2\sqrt{t})} \right)$ $\frac{dy}{dx} = -\frac{1}{2} \tan(2\sqrt{t})$		
6(b) (ii)	$\frac{d^2y}{dx^2} = \left[-\frac{1}{2} \sec^2(2\sqrt{t}) \left[2 \left(\frac{1}{2\sqrt{t}} \right) \right] \right] \times \frac{\sqrt{t}}{\cos(2\sqrt{t})}$ $= -\frac{\sec^2(2\sqrt{t})}{2 \cos(2\sqrt{t})}$ $= -\frac{1}{2} \sec^3(2\sqrt{t})$ When $t = \frac{\pi}{4}$: $\frac{d^2y}{dx^2} = -\frac{1}{2} \sec^3 \left(2\sqrt{\frac{\pi}{4}} \right)$ $= -\frac{1}{2 \cos^3(\sqrt{\pi})}$ $= 62.2$		Differentiate (ignore $\times \frac{dt}{dx}$) Simplify (must $\times \frac{dt}{dx}$) 1 decimal place
TOTAL MARKS		15 marks	

NO	ANSWER SCHEME	MARK	NOTES
7(a)	$\frac{dV}{dt} = 500$ $\frac{r}{h} = \frac{400}{1600} = \frac{1}{4}$ $r = \frac{1}{4}h$ $V = \frac{1}{3}\pi r^2 h$ $V = \frac{1}{3}\pi \left(\frac{1}{4}h\right)^2 h = \frac{1}{48}\pi h^3$ $\frac{dV}{dh} = \frac{1}{48}\pi(3h^2)$ $\frac{dV}{dh} = \frac{\pi h^2}{16}$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $\frac{dh}{dt} = \frac{16}{\pi h^2} \times 500$ $\frac{dh}{dt} = \frac{8000}{\pi h^2}$		Express r in terms of h Express V in terms of h Differentiate his V Correct $\frac{dV}{dh}$

NO	ANSWER SCHEME	MARK	NOTES
7(b)	$r = \frac{1}{4}h$ $\frac{dr}{dh} = \frac{1}{4}$ When $r = 200$: $h = 4(200) = 800$ $\frac{dr}{dt} = \frac{dr}{dh} \times \frac{dh}{dt}$ $\frac{dr}{dt} = \frac{1}{4} \times \frac{8000}{\pi(800)^2}$ $\frac{dr}{dt} = \frac{1}{320\pi} \text{ cms}^{-1}$		Find $\frac{dr}{dh}$ Find h (must use $r = 200$)
TOTAL MARKS		10 marks	